

FieldAI - Combined BRD & PRD Specification

Business Requirements Document (BRD) & Product Requirements Document (PRD)

Document Type:	Unified BRD & PRD Specification	Product Stage:	Production Baseline v1.0
System Identifier:	FieldAI Audit Assistant	Target Environment:	Render Cloud (Docker + PostgreSQL)
Orchestration:	LangGraph Multi-Agent StateGraph	AI Engine:	DeepSeek Flash & ElevenLabs Scribe
Author / Practice:	Sami Associates Audit Technology	Product Approver:	Product Owner (Sami)

Executive Overview & Purpose: This unified specification bridges business requirements (BRD) with technical and architectural execution (PRD) for FieldAI, the enterprise internal audit fieldwork platform. FieldAI transforms raw audit walkthrough recordings, multi-lingual interviews, and auditee procedural documentation (SOPs, manuals, policies) into a fully linked, verifiable, and standardized audit workpaper ecosystem. The system coordinates 21 specialized LangGraph agents across 6 structured sub-graphs, an interactive 15-page Streamlit portal, an asynchronous PostgreSQL job queue worker, and deterministic audit analytics logic to cut audit cycle times by over 50% while guaranteeing 100% bi-directional traceability back to source timestamps and document clauses.

Part I: Business Requirements Document (BRD)

1.1 Business Problems & Market Drivers

Internal audit departments face severe operational constraints during fieldwork and walkthrough phases: (1) Manual Documentation Friction: Auditors spend 40-60% of their fieldwork hours transcribing interviews, drafting step narratives, and manually drawing flowcharts. (2) Information Asymmetry: What auditees say in walkthroughs frequently diverges from what is written in approved SOPs, creating unobserved control gaps. (3) Broken Traceability: In standard workpapers, audit test steps rarely maintain direct evidence links to the exact interview timestamp or policy clause. (4) Audit Standard Rigor: The IIA Global Internal Audit Standards (2024 Edition) mandate thorough evidence grounding, supervisory review, and robust quality assurance.

1.2 Strategic Business Objectives & Quantitative ROI

Goal ID	Strategic Objective	Target Metric / Success Threshold	Business Impact
G-01	Accelerate Walkthrough Lead Time	>= 50% reduction in elapsed time from interview to approved RCM.	Frees senior auditors from administrative drafting to focus on high-risk testing.
G-02	Single Source of Truth	Zero manual re-keying across Process Table, Flowcharts, RCM, and Program.	Eliminates version mismatch and inconsistent risk-control numbering.
G-03	Bi-Directional Source Traceability	100% of steps, risks, controls, and tests cite exact timestamp or clause.	Defensible audit trails during external quality assessment reviews (EQAR).
G-04	Objective Gap & Divergence Detection	100% automated 5-category gap reconciliation between SOPs and interviews.	Uncovers informal or undocumented operational workarounds immediately.
G-05	Full-Population Substantive Testing	100% verification of duplicate payments, split POs, and SoD toxic roles.	Transitions audits from small sample sizes (25 items) to full population analytics.

1.3 Stakeholder Personas & Permissions (RBAC)

- Chief Audit Executive (CAE): Reviews audit committee dashboards, process risk heatmaps, control gap densities, and continuous monitoring exceptions.
- Audit Manager: Reviews change sets, resolves contradictory walkthrough testimonies, approves RCM design ratings, and formally locks audit test programs.
- Field Senior / Auditor: Records/uploads walkthrough audio, inspects transcribed segments, edits process tables, reviews visual flowcharts, and runs substantive tests.
- Auditee / Business Process Owner: Receives token-secured, read-only links via the confirmation portal to inspect visual swimlanes and confirm process accuracy.
- Audit System Administrator: Manages user credentials, sampling parameter tables, risk/control libraries, regulatory framework mappings, and AI API keys.

1.4 Regulatory & Framework Alignment

FieldAI maps all operational controls directly to international and regional governance frameworks: IIA Global Internal Audit Standards (2024), COSO Internal Control - Integrated Framework (2013), COBIT 2019, ISO 27001:2022, Saudi National Cybersecurity Authority (NCA ECC), and SAMA Cybersecurity Framework (CSF).

Part II: Product Requirements Document (PRD)

2.1 System Architecture & Multi-Tier Cloud Topology

FieldAI operates as a modular, decoupled containerized platform on Render Cloud, specified via render.yaml:

Service	Type	Technology	Architecture & Responsibility
fieldai-web	Web Service	Streamlit 1.35+, Docker	Serves the 15-module auditor portal, st.data_editor, Cytoscape.js canvas, and st.audio_input.
fieldai-worker	Background Worker	worker.py, Python 3.11	Polls PostgreSQL jobs table via SELECT ... FOR UPDATE SKIP LOCKED. Executes transcription, OCR, and mining.
fieldai-monitoring	Cron Job	run_monitoring.py, Cron	Executes weekly scheduled continuous monitoring scripts against new client data extracts.
fieldai-db	Managed DB	PostgreSQL 16, psycpg3	Single store: Master models, version snapshots, immutable audit trail, files (BYTEA), and LangGraph checkpoints.
External AI	SaaS APIs	ElevenLabs & DeepSeek	ElevenLabs Scribe (word timestamps + diarization) and DeepSeek Flash for structured reasoning.

2.2 Visualization & Deliverables Engine

FieldAI provides a specialized dual-engine visualization architecture designed to handle complex audit flowcharts with non-bold Inter typography, curved connectors, swimlane segregation, and high-density information compacting:

- Cytoscape.js Interactive Canvas (Dagre + Bezier Routing): Employs the Dagre hierarchical layout algorithm with top-to-bottom orientation (rankDir: 'TB'), tight node/rank separation (nodeSep: 35, rankSep: 45), and automatic compound bounding for swimlanes. Features smooth curved Bezier routing, draggable nodes, click-to-inspect audit drawer, density presets (Micro, Compact, Full), and PNG/JSON export.
- High-Fidelity Graphviz Architecture Blueprint: Rendered at authentic 1:1 pixel scale (use_container_width=False) to prevent magnification distortion. Features smooth spline routing, compact pill terminals, proportional decision diamonds, and left-aligned swimlane cluster labels.

Standard Deliverable Export Matrix:

Export Format	Extension	Underlying Generator	Target Application & Interoperability
BPMN 2.0 XML	.bpmn	exports/bpmn.py	Standard OMG BPMN 2.0 XML with full swimlanes, gateway nodes, and sequence flows for Camunda and enterprise BPM suites.
draw.io / Diagrams	.drawio	exports/drawio.py	Native draw.io XML with swimlane containers and connected vertices. Can be directly converted to Microsoft Visio (.vsdx).
Audit Summary PDF	.pdf	exports/pdf.py (ReportLab)	Publication-quality formal deliverable with Inter typography, step tables, and risk/control listings.
Graphviz DOT	.dot	agents/flowchart_agent.py	Raw Graphviz source code formatted with cluster swimlanes for automated CI/CD documentation pipelines.
Cytoscape JSON	.json	analytics/flowchart_viewer.py	Complete node/edge element graph and layout metadata for web-based graph visualization engines.

2.3 Comprehensive Application Screen Catalogue (15 Modules)

The web user interface is partitioned into 15 purpose-built Streamlit modules under pages/, designed specifically for audit teams to progress systematically from engagement setup to fieldwork delivery:

File & Module	Key Functional Capabilities & Auditor Workflows
app.py / Portal	Bcrypt login authentication, role-based navigation bar, active engagement context switcher, and global Ask FieldAI semantic copilot.
1_Engagements.py	Create/manage audit engagements, define process entities, assign stable code prefixes (e.g. P2P, O2C, HTR), and monitor fieldwork progress.
2_Capture.py	Direct browser audio recording via st.audio_input, multi-format upload (MP3, WAV, M4A, MP4, OGG), mandatory consent logging, diarization, speaker role mapping, segment redaction, and near-live interview copilot.
3_Review_Changes.py	Multi-meeting diff analysis; tracked changes review (Added, Changed, Confirmed, Contradicted, Removed); side-by-side conflict resolution; mandatory resolution notes; and version bumping (v1.0 -> v1.1).
4_Process_Table.py	Interactive master model table with in-place st.data_editor; columns for Step Code, Order, Description, Role, Department, System, Decision Gates, and source timestamps with confidence ratings.
5_Flowchart.py	Dual-engine visualization: Cytoscape.js interactive canvas (Dagre layout, Bezier curved connectors, draggable nodes, density modes: Micro/Compact/Full, click-to-inspect audit drawer) and Graphviz Blueprint (1:1 scale). 5 export formats.
6_RCM.py	Risk-Control Matrix with inherent risk ratings, control design adequacy evaluations, AI-suggested controls for unmitigated gaps, framework mapping (COSO, COBIT, ISO, NCA, SAMAs), and Excel import/export.
7_Audit_Program.py	Test of Design (ToD) and Operating Effectiveness (ToE) procedures; deterministic sample size calculation from sampling table; PBC request lists; criteria references; manager approval and workpaper lock.
8_Documents.py	Document metadata register; multi-format upload (PDF OCR, DOCX, XLSX, PPTX); 5-category reconciliation against walkthrough interviews (Matches, Documented-not-described, Described-not-documented, Conflicting, Outdated); document citation Q&A.;
9_Prep.py	Walkthrough preparation engine generating bilingual (Arabic and English) interview question packs grouped by step, follow-up interview agendas, and risk-based scoping matrices.
10_Testing.py	5 substantive testing engines: Automated Analytics (duplicate payments, split POs, Benford's Law), Segregation of Duties (SoD) toxic role matrices, Process Mining conformance, Evidence OCR attribute verification, and Recurring Tests.
11_Findings.py	Drafts formal audit observations in the 5 Cs format (Condition, Criteria, Cause, Effect, Recommendation); assigns risk ratings; and evaluates workpapers against IIA Global Standards 2024 quality checklist.
12_Dashboard.py	Executive CAE metrics, process risk heatmaps, control gap density distributions, fieldwork completion rates, and real-time exception alert feeds.
13_Confirm.py	Token-secured external portal enabling auditees to view approved process flows, inspect visual swimlanes, and submit validation comments or corrections.
14_Admin.py	User management & RBAC, sampling parameter tables, risk & control reference libraries, AI model credentials, and immutable audit trail viewer.
15_Centralized_Storage.py	Centralized enterprise evidence repository: folder tree hierarchy, multi-tag filtering, metadata indexing, document preview, secure downloads, and storage governance.

Part III: Multi-Agent Architecture, Configuration & Flow

FieldAI executes intelligence through 21 autonomous, single-responsibility agents in agents/, orchestrated via LangGraph StateGraph. State transitions, conditional branches, and human-in-the-loop review interrupts are typed under FieldAIState (graphs/state.py).

3.1 Agent Configuration Matrix (All 21 Agents)

Agent Name	Engine / Model	Core Mission & Logic	Input -> Output Contract
transcription_agent	ElevenLabs Scribe	Speech-to-text with word timestamps, speaker diarization, language auto-detection (AR/EN), and redaction.	audio file -> transcript_segments
summary_agent	DeepSeek Flash	Synthesizes walkthrough narrative into executive summary, action items, open questions, and initial PBC requests.	transcript -> summary, open_items, pbc
process_extraction_agent	DeepSeek Flash	Extracts ordered steps, responsible roles, departments, systems, inputs/outputs, and decision diamonds.	transcript -> extracted_steps
risk_control_agent	DeepSeek Flash	Identifies stated and library-suggested risks and controls, categorizes nature/frequency/owner, and flags initial SoD gaps.	steps -> extracted_risks, extracted_controls
change_set_agent	DeepSeek Flash	Differs new extraction against master model. Resolves entities, detects contradictions, and pauses for human review.	new vs master -> change_set (Interrupt)
flowchart_agent	Deterministic	Constructs Graphviz DOT and Cytoscape JSON graphs with swimlane clustering, risk/control badges, and compact styling.	master model -> dot_code, cy_elements
rcm_agent	DeepSeek Flash	Builds RCM rows, rates control design adequacy (Adequate/Partially/Inadequate), and maps regulatory frameworks.	model -> rcm_rows, design_ratings
audit_program_agent	DeepSeek + Rules	Formulates ToD and ToE steps, calculates deterministic sample sizes from sampling table, and sets PBC evidence lists.	rcm -> test_steps, sample_sizes
document_agent	DeepSeek + OCR	Parses auditee SOPs, policies, manuals (PDF/DOCX/XLSX), extracts metadata, and chunks text with section citations.	document file -> doc_chunks, doc_metadata
reconciliation_agent	DeepSeek Flash	Performs 5-category gap analysis comparing documented procedures with walkthrough interview statements.	doc vs interview -> recon_items
prep_agent	DeepSeek Flash	Generates bilingual walkthrough interview question packs, follow-up agendas, and risk-based scoping matrices.	prior model -> bilingual question pack
copilot_agent	Scribe + DeepSeek	Near-live walkthrough assistant analyzing audio chunks to propose real-time probing questions and instant PBC items.	audio chunk -> live_prompts, instant_pbc
analytics_agent	Pandas + Rules	Maps uploaded data schemas and runs fixed analytics: duplicate payments, split POs, weekend postings, Benford's Law.	client data extract -> test_exceptions
sod_agent	Rules + DeepSeek	Evaluates user-access matrices against toxic role combination rules, mapping conflicts to swimlane handoffs.	user permissions -> sod_conflicts
process_mining_agent	Pandas + DeepSeek	Analyzes event logs to identify process path variants, cycle time bottlenecks, and control bypasses.	event log -> variants, bypass_alerts
evidence_agent	OCR + DeepSeek	Inspects sample vouchers (PO, invoice, approval email) and verifies compliance with audit test attributes.	voucher files -> attribute_results
monitoring_agent	Deterministic	Schedules recurring automated tests, logs periodic cron runs, and raises exception alerts on new data.	cron trigger -> updated_exceptions
findings_agent	DeepSeek Flash	Drafts formal audit observations in the 5 Cs format (Condition, Criteria, Cause, Effect, Recommendation).	exceptions -> structured 5 Cs findings
qa_review_agent	DeepSeek Flash	Reviews working papers against IIA Global Standards 2024 checklist and methodology rules, creating review notes.	workpapers -> qa_checklist, notes
knowledge_agent	DeepSeek Flash	Enables cross-engagement intelligence reuse, baseline process templates, and year-on-year change detection.	prior engagements -> baseline_model
doc_qa_agent	DeepSeek Flash	Performs semantic search across transcripts and documents, answering auditor queries with exact citations.	user question -> grounded cited answers

3.2 Detailed Agentic Workflows & State Transitions

FieldAI executes intelligence through 6 distinct StateGraph pipelines. Each graph coordinates node executions, passes typed state (FieldAIState), and handles human review checkpoints.

Workflow 1: Core Walkthrough Meeting Graph (graphs/meeting_graph.py)

Seq	State Node / Agent	Input Context	Processing Logic & Verification Guardrail	State Output Artifact
1	transcription (transcription_agent)	Audio file (files table)	Checks consent_recorded. Calls ElevenLabs Scribe (fallback OpenAI). Segments speech by speaker turn with second-level timestamps.	transcript_segments
2	summary (summary_agent)	transcript_segments	DeepSeek Flash synthesizes <=200 word summary, key decisions, opens items, and PBC requests.	summary, open_items, pbc_requests
3	process_extraction (process_extraction_agent)	transcript_segments	Extracts ordered steps, owners, systems, decision gates, inputs/outputs. Assigns stable step codes (e.g. P2P-01).	extracted_steps (ProcessStep[])
4	risk_control (risk_control_agent)	extracted_steps	Identifies stated auditee risks and controls. Proposes library controls for unmitigated gap risks. Flags SoD conflicts on handoffs.	extracted_risks, extracted_controls
5	change_set (change_set_agent)	extractions vs master model	Diff extraction against current master model. Classifies items: Added, Changed, Confirmed, Contradicted, Removed.	change_set, conflict_list
6	HUMAN REVIEW INTERRUPT (LangGraph interrupt())	change_set	Execution halts; PostgresSaver checkpointer stores thread state. Auditor reviews tracked changes and resolves conflicts on Review Changes page.	review_decisions (resume)
7	apply_changes (core/model_repo.py)	review_decisions	Applies approved deltas to master model tables (steps, risks, controls). Bumps version snapshot (v1.0 -> v1.1) in versions table.	updated master model
8	flowchart (flowchart_agent)	master model	Deterministic generator outputs Graphviz DOT and Cytoscape JSON with lane clustering and density modes (Micro, Compact, Full).	dot_code, cy_elements
9	rcm (rcm_agent)	master model	Builds formal RCM, rates control design adequacy, and maps regulatory standards (COSO, COBIT, ISO, NCA, SAMA).	rcm_rows, design_ratings
10	audit_program (audit_program_agent)	rcm_rows	Drafts ToD and ToE test procedures, calculates deterministic sample sizes from sampling table, and sets PBC evidence lists.	test_steps, sample_sizes

Workflow 2: Document Ingestion & Reconciliation Graph (graphs/document_graph.py)

Seq	State Node / Agent	Input Context	Processing Logic & Verification Guardrail	State Output Artifact
1	document (document_agent)	Policy / SOP file (PDF OCR, DOCX, XLSX)	Extracts document register metadata (owner, version, effective date, approval status). Chunks document into section/page segments with citations.	doc_chunks, doc_metadata
2	reconciliation (reconciliation_agent)	doc_chunks vs interview master model	Executes 5-Category Gap Analysis comparing written procedures vs interview testimony: (1) Matches, (2) Documented-not-described, (3) Described-not-documented, (4) Conflicting detail, (5) Outdated document.	recon_items, gap_notes
3	HUMAN REVIEW INTERRUPT (Documents Page)	recon_items	Auditor inspects reconciliation findings side-by-side with SOP text, confirms criteria references, and selects items to merge into master model.	accepted_recon_items
4	rebuild_deliverables (flowchart -> rcm -> audit_program)	accepted_recon_items	Re-runs deliverables graph to attach verified SOP clause citations to control criteria columns and refresh flowcharts.	updated deliverables

Workflow 3: Substantive Field Testing Graph (graphs/testing_graph.py)

Branch / Node	Executing Agent	Data Input Source	Deterministic Testing Logic & Exceptions Detected	Deliverable Artifact
Router	test_router (conditional edge)	test_request.test_type	Evaluates requested test category and dynamically routes execution state to analytics, sod, process_mining, or evidence.	Branch routing dispatch
Branch A: Analytics	analytics_agent (Pandas library)	Client financial extract (CSV / Excel)	Full-population queries: duplicate payments, split POs below approval limits, weekend postings, round amounts, Benford's Law digit tests, 3-way match exceptions.	test_results, exception_rows
Branch B: SoD	sod_agent (Matrix engine)	ERP user-role-permission dump	Evaluates permission matrix against toxic role pairs (e.g. create vendor + post payment, record invoice + approve payment). Maps conflicts to swimlane handoffs.	sod_conflicts, violation_matrix
Branch C: Mining	process_mining_agent (Event log engine)	Audit trail event log (case_id, activity, time, user)	Computes process variant frequencies, throughput cycle times, happy path deviation ratios, and flags transactions that bypassed key controls.	variants, bypass_alerts
Branch D: Evidence	evidence_agent (OCR attribute matcher)	Sample voucher files (PO, invoice, approval email)	OCR-inspects voucher files, verifies audit test attributes (authorized signature, amount matching PO, valid tax invoice), and quotes verified text.	attribute_results, quotes
Recurring Check	check_recurring -> monitoring_agent	make_recurring flag == True	If enabled, registers test in test_runs table as recurring. Weekly Render Cron job (run_monitoring.py) re-executes tests on new data extracts.	active_recurring_tests

Workflows 4, 5 & 6: Reporting, Preparation, and Global Orchestrator

Workflow Pipeline	Trigger & Agents	State Progression & Operational Behavior
Reporting & QA (graphs/reporting_graph.py)	findings_agent -> qa_review_agent	Triggered from Findings Page. findings_agent synthesizes test exceptions and reconciliation gaps into formal 5 Cs audit observations (Condition, Criteria, Cause, Effect, Recommendation). qa_review_agent evaluates the complete audit file against IIA Global Standards 2024 checklist.
Meeting Prep & Copilot (graphs/prep_graph.py)	prep_agent -> copilot_agent	Triggered from Prep and Capture Pages. prep_agent analyzes prior year RCMs and uploaded SOPs to generate bilingual (Arabic and English) interview question packs. copilot_agent analyzes near-live 2-5 minute audio chunks, outputting real-time probing questions.
Global Orchestrator (graphs/orchestrator.py)	Supervisor Router + doc_qa_agent	Central dispatcher: Routes UI actions deterministically to sub-graphs without LLM cost. Free-text queries from 'Ask FieldAI' box are classified into target workflows or answered with exact citations via doc_qa_agent across transcripts and documents.

Part IV: Data Architecture, Security & Governance

4.1 Relational Data Model (PostgreSQL 16)

Table Name	Primary / Foreign Keys	Key Data Attributes & Schemas	Storage Purpose & Audit Role
users	id (PK)	username, password_hash, role, active, created_at	User credentials with bcrypt cryptographic hashing and RBAC.
engagements	id (PK)	name, entity, period, status, created_at	Top-level audit containers (e.g. FY26 IT Audit).
processes	id (PK), engagement_id (FK)	name, code_prefix, current_version, status	Process definitions with stable code prefixes (P2P, O2C).
sources	id (PK), process_id (FK)	kind, title, storage_path, language, consent_recorded	Evidence registry; transcription blocked without consent.
transcript_segments	id (PK), source_id (FK)	speaker_label, speaker_name, start_s, end_s, text, redacted	Diarized utterances with exact second timestamps.
steps	id (PK), process_id (FK)	step_code, order_num, description, role, system, is_decision	Master flow steps. Never renumbered; status='withdrawn'.
risks	id (PK), process_id (FK)	risk_code, description, inherent_rating, ai_suggested	Identified inherent risk catalogue with severity scoring.
controls	id (PK), process_id (FK)	control_code, description, type, nature, freq, key_control	Internal controls with design adequacy ratings & framework refs.
versions	id (PK), process_id (FK)	version_label, snapshot_json, change_log, approved_by	Immutable version snapshots (v1.0, v1.1) approved by managers.
change_sets	id (PK), process_id (FK)	source_id, status (proposed/applied), changes_json	Multi-meeting delta proposals awaiting human auditor review.
documents	id (PK), source_id (FK)	doc_type, owner, version, effective_date, approval_status	Auditee document metadata register with freshness tracking.
recon_items	id (PK), process_id (FK)	category, item_code, doc_ref, meeting_ref, note, status	5-category reconciliation items between SOPs and interviews.
tests	id (PK), control_id (FK)	test_code, objective, test_type, procedure, sample_size	Audit program test steps linked to deterministic sampling rules.
test_runs	id (PK), test_id (FK)	run_type, parameters_json, result_summary, exceptions_json	Substantive test executions and continuous monitoring log.
findings	id (PK), process_id (FK)	finding_code, condition, criteria, cause, effect, recommendation	5 Cs formal audit observations linked to controls and tests.
files	id (PK)	filename, mime_type, size_bytes, data (BYTEA), sha256	Centralized binary storage accessible to web and worker.
jobs	id (PK)	job_type, payload_json, status, progress, message, thread_id	Asynchronous worker task queue (SELECT ... FOR UPDATE SKIP LOCKED).
audit_log	id (PK)	user_id, action, entity, entity_id, details_json, created_at	Append-only immutable audit trail recording all user actions.

4.2 Non-Functional Requirements & Security Governance

- Zero Arbitrary AI Code Execution: To guarantee audit integrity, no LLM-generated code is ever executed against client financial data. All substantive testing relies on verified, deterministically coded Python/Pandas functions in analytics/library.py.
- Auditor-in-the-Loop Safeguards: All AI extractions are treated as draft recommendations. Human auditors retain full override authority. Changes to key controls require explicit managerial re-approval before updating master baselines.
- Bi-Directional Source Traceability: 100% of steps, risks, controls, tests, and findings maintain clickable links to raw audio timestamps (e.g. 00:14:22) or document section citations (e.g. SOP v2.1 Section 4.2).
- Privacy & Redaction: Mandatory consent logging blocks transcription until consent is verified. Auditors can redact sensitive PII or off-the-record statements prior to extraction.
- Immutable Audit Trail: Every login, model edit, change-set approval, and export action is permanently recorded in the append-only audit_log table with SHA-256 hash verification.

4.3 Acceptance Verification & Stakeholder Sign-Off

Prepared by: Lead Audit Solutions Architect

Signature: _____

Date: October 2026

Approved by: Product Owner & Practice Lead (Sami)

Signature: _____

Status: Approved Production Specification v1.0